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ÉTUDES INTERNATIONALES
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GRADUATE INSTITUTE
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DEVELOPMENT STUDIES

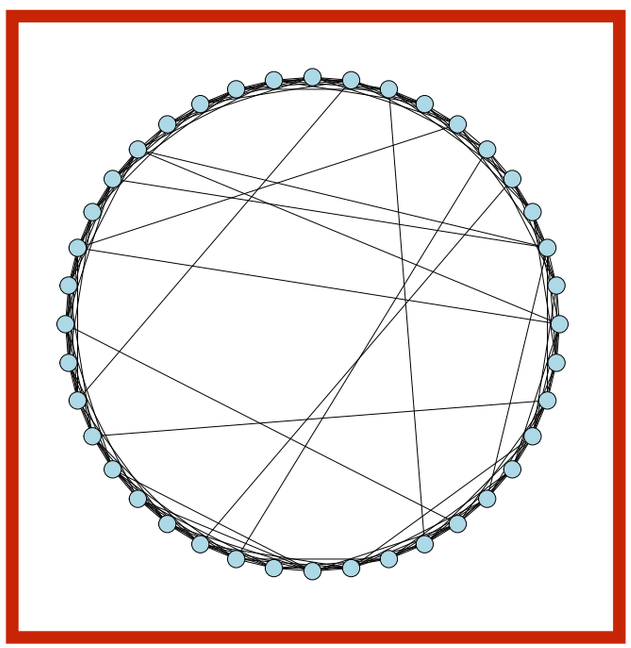
# Topology

## Social Networks Theory and Methods

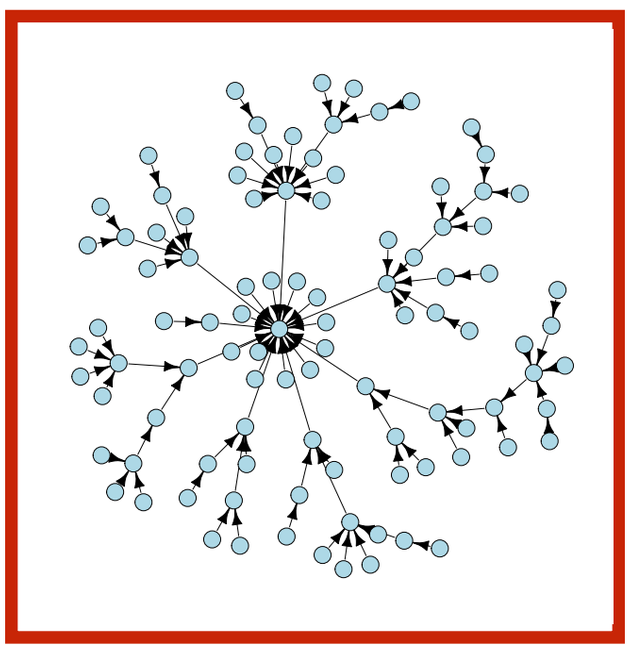
James Hollway

# Topology

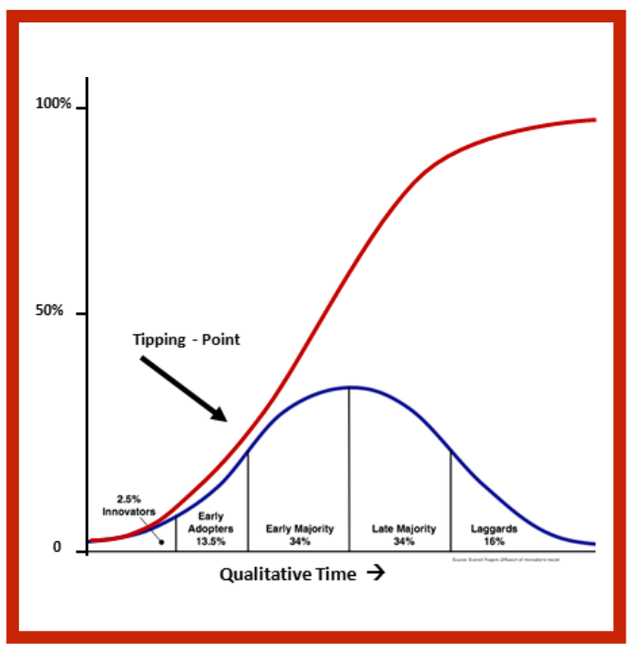
## Small-World



## Scale-Free

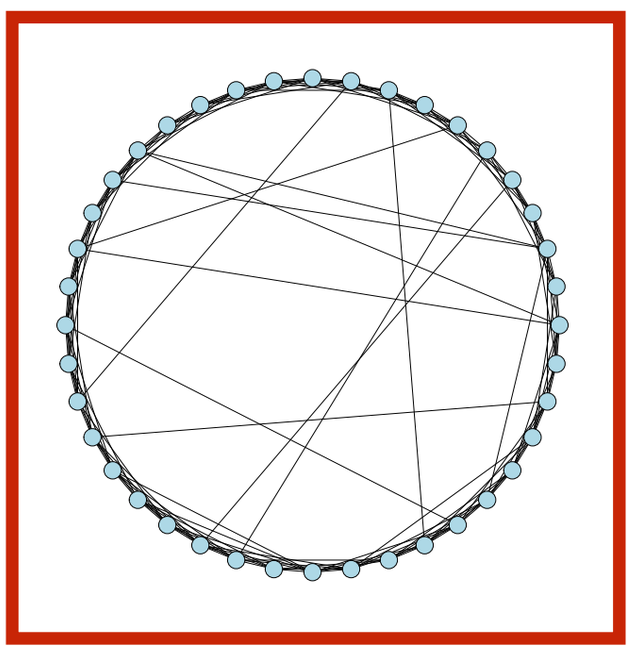


## Diffusion



# Topology

## Small-World



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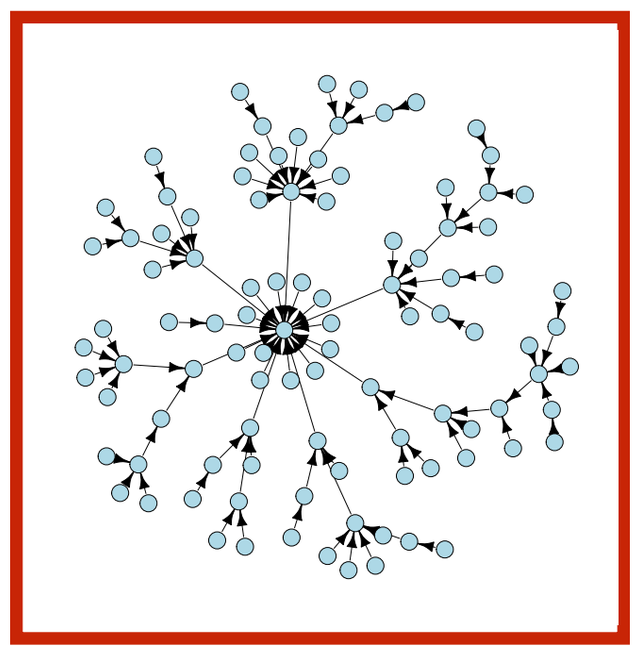

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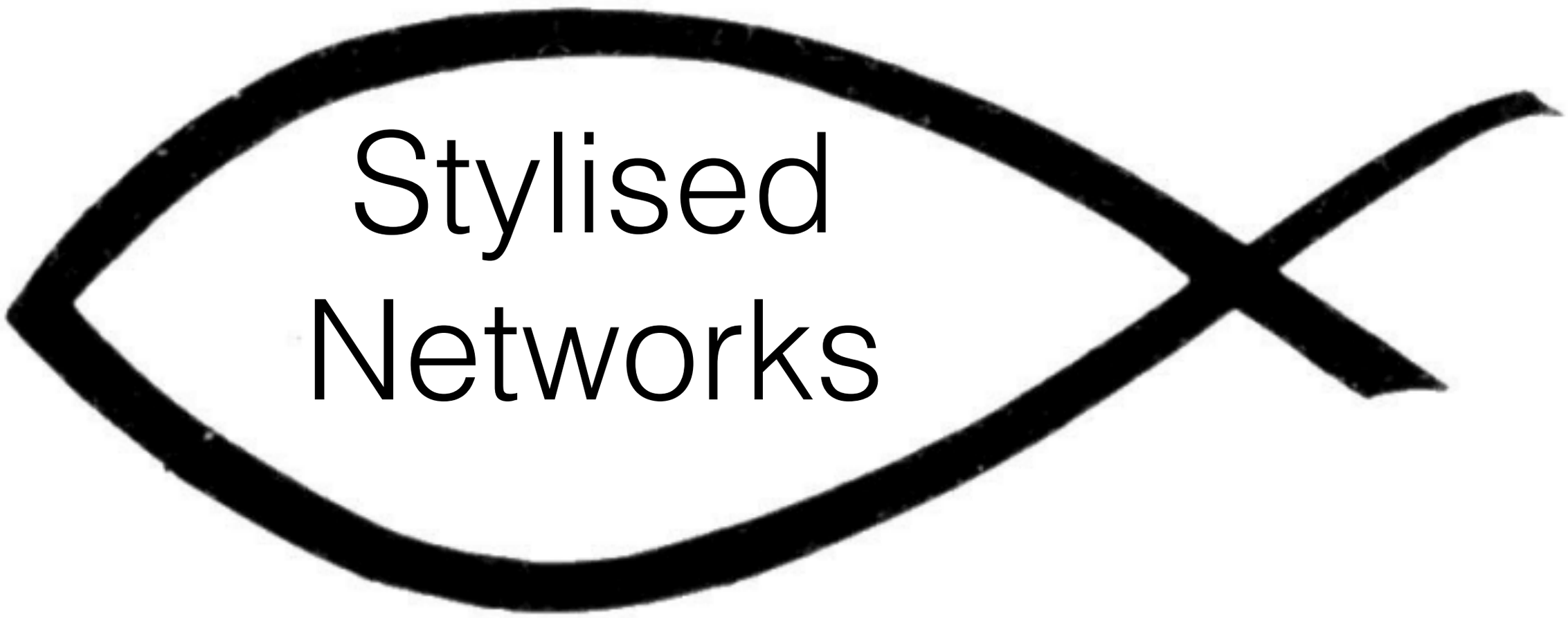
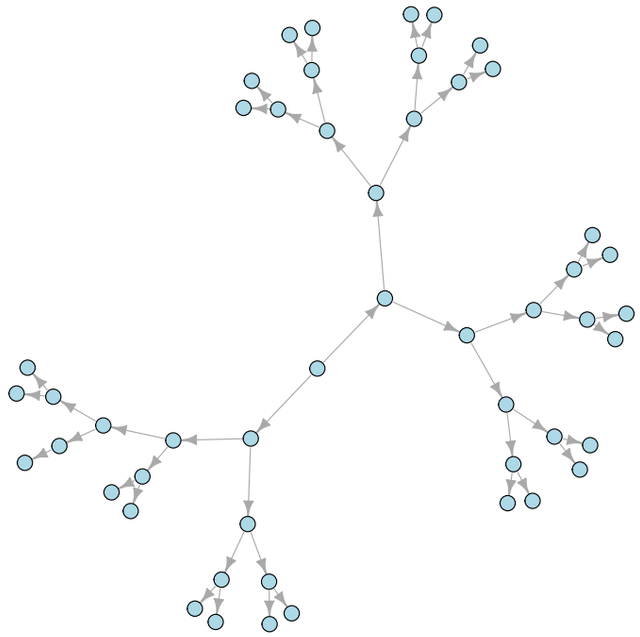

Stylised
Networks

# What are stylised network models?

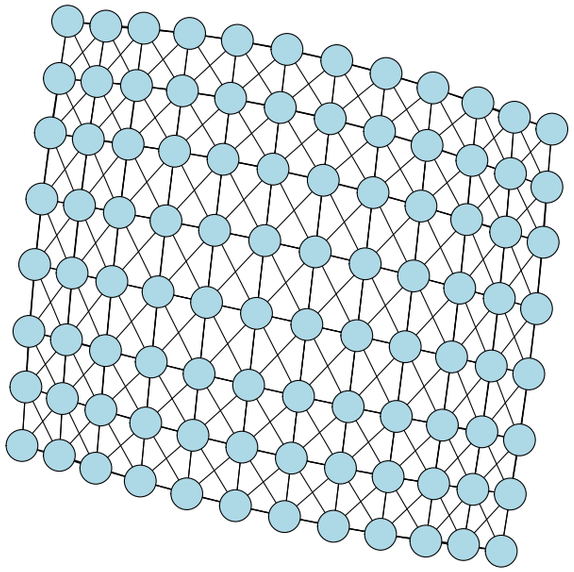
- Ideal type networks
  - Constructed according to 1-2 simple rules
  - Exaggerate structural features commonly found in networks
    - Centrality
    - Cohesion
    - Randomness

# 1. Regular Tree



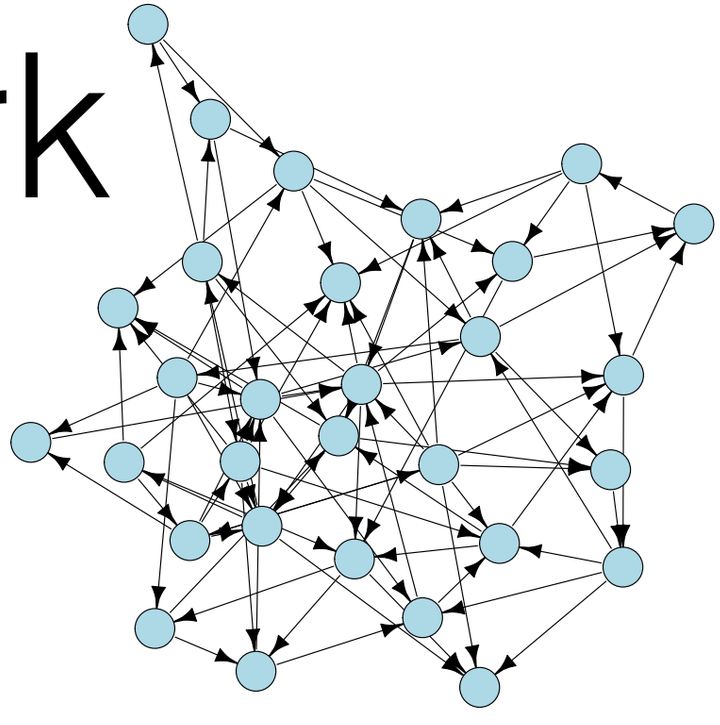
- Rationale:
  - Some networks tend to be centralised, i.e. some nodes have better reach than others and/or network is asymmetric
  - Often the case in asymmetric, functional, or hierarchically organised settings
- Generated by creating a network of *branching* nodes with parameters:
  - Number of branches per node (here 2)
  - Network distance (generations) or dimensions (here $n$=50)
- Uses: some use in organizational and biological networks

# 2. Regular Lattice



- Rationale:
  - Some networks tend to be clustered, i.e. there is a high likelihood that one's interaction partners also interact
  - Often the case where geographic or social space important
- Generated by arranging nodes on a lattice with parameters:
  - Can vary by number of dimensions (here 2)
  - And neighbourhood/interaction distance (here 1+diagonal)
- Uses: commonly used in ABM to show how spatial or network clustering can allow or limit diffusion (more later) or make pockets of behaviour stable (more next week)

# 3. Random network



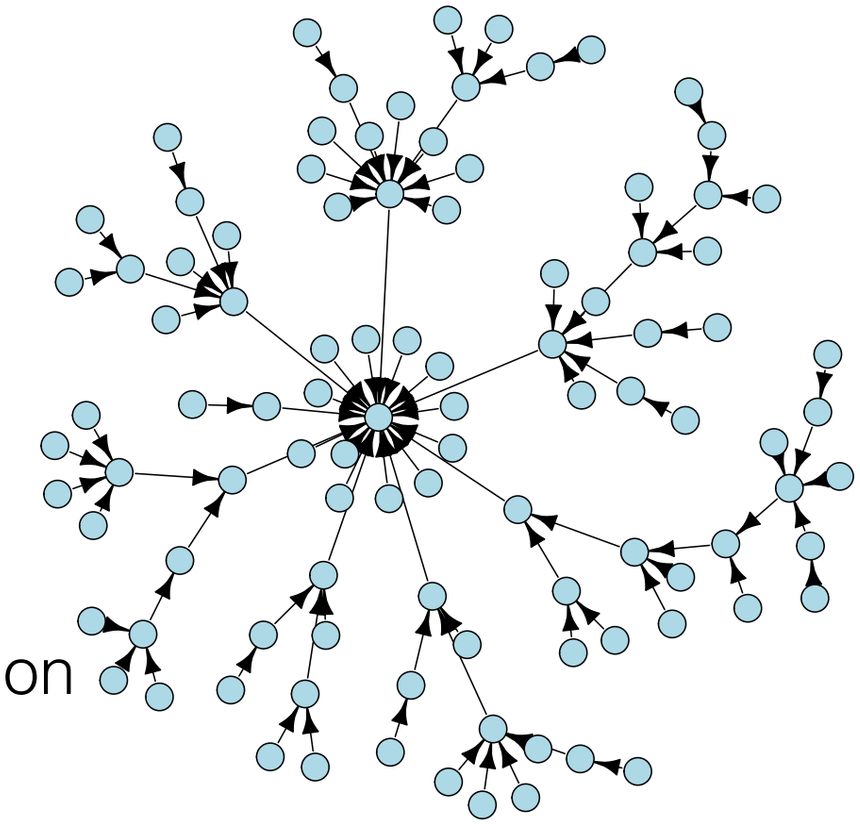
- Also known as a Bernoulli network (after Swiss mathematician Jacob Bernoulli, brother to Johann, Euler's advisor) or an Erdös-Renyi (1959; see also Rapaport 1953) network

- Rationale:
  - Some networks tend to be random, i.e. where each tie has an equal probability of existing
  - Rarely the case, but used as below

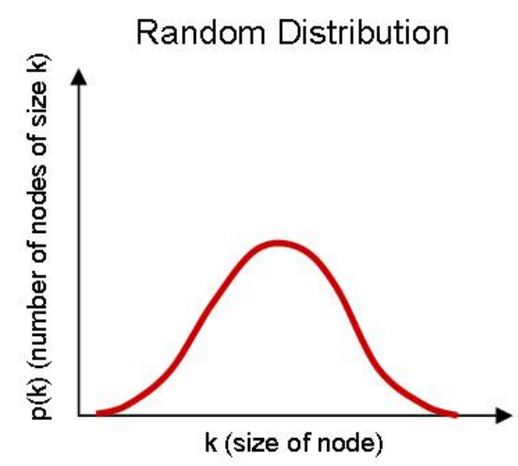
- Generated by creating edges at random on a network with parameters:
  - Nodes (here 30)
  - Density

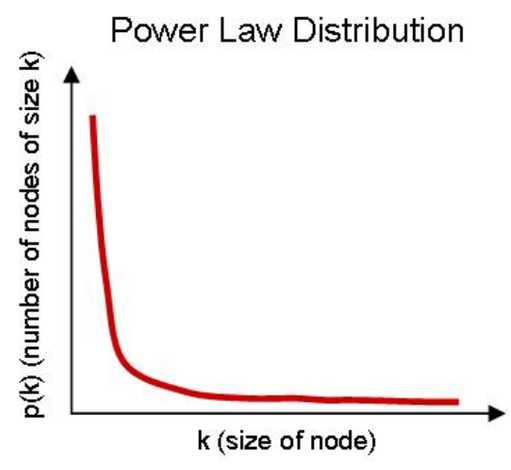
- Uses: often used as a simple baseline to ascertain whether a certain substructure observed more often than expected by chance

# 4. Scale Free



- Rationale:
  - Some networks have a strong degree dispersion
  - Often the case where positive feedback mechanisms prevalent (e.g. internet, twitter, academic papers)
- Generated by iterative creation of a network where each new node attaches to existing nodes with probability proportional to their degree
  - Degree distribution of this generative process follows a power law
- Uses: some scholars claim power laws are common feature of many networks



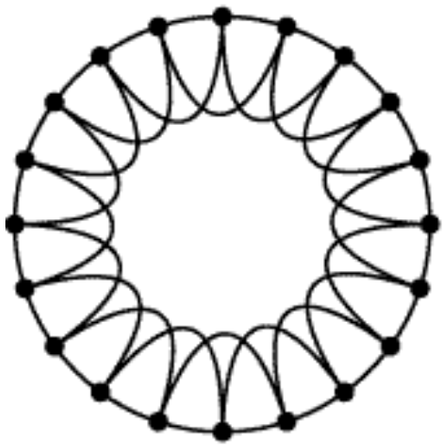
# The Matthew Effect

- In 1968, Robert Merton published an article called “The Matthew Effect in Science”
  - “For unto every one that hath shall be given, and he shall have abundance; but from him that hath not shall be taken even that which he hath” Matthew 25:29
- In this article, he discussed how academic fame leads to more fame — in terms of prizes, citations, and attribution of merit (cumulative advantage or preferential attachment)
- In citation networks this mechanism in its purest form can lead to so-called scale-free networks

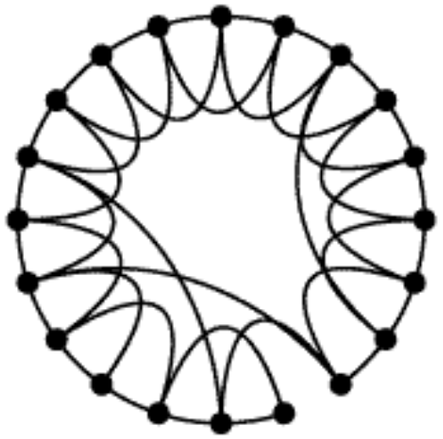
# 5. Small World

- Rationale
  - Some networks tend to be clustered by interconnected by just a few spanning ties
  - Often the case across a surprising range of settings
- Generated by creating a (ring) lattice and then rewiring a few ties at random with parameters:
  - Lattice dimensions and distance
  - Probability of a tie being rewired
- Uses: commonly used to show “it’s a small world after all” and to model diffusion

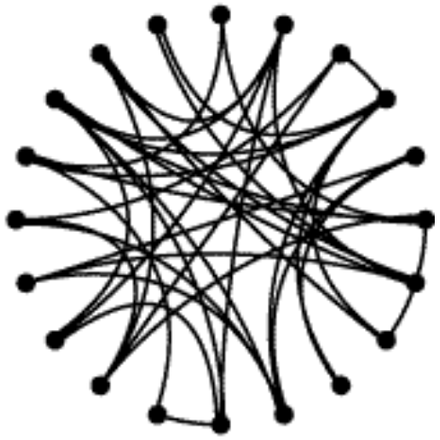
Regular



Small-world



Random



# The Milgram Experiment

- In the 60s, Stanley Milgram did an experiment where he invited (through advertisement) people to send a letter to a person unknown to them through intermediaries

- He found that everybody was connected to everyone else through 6 steps
  - He thought this was evidence we lived in a “small world”

- What was wrong with his experiment?

# What is wrong with Milgram's (small) world?

- Selection bias: "starters" were recruited through an advertisement searching for well-connected people

- Non-response bias: if one assumes an attrition rate, longer chains will be underrepresented

- Greedy algorithm: people can only make local decisions and cannot omnisciently recognise the shortest global path

- But, Watts and Strogatz (1998) showed this using computer simulations and rise of relational rewiring has brought "6 degrees of separation" into popular culture

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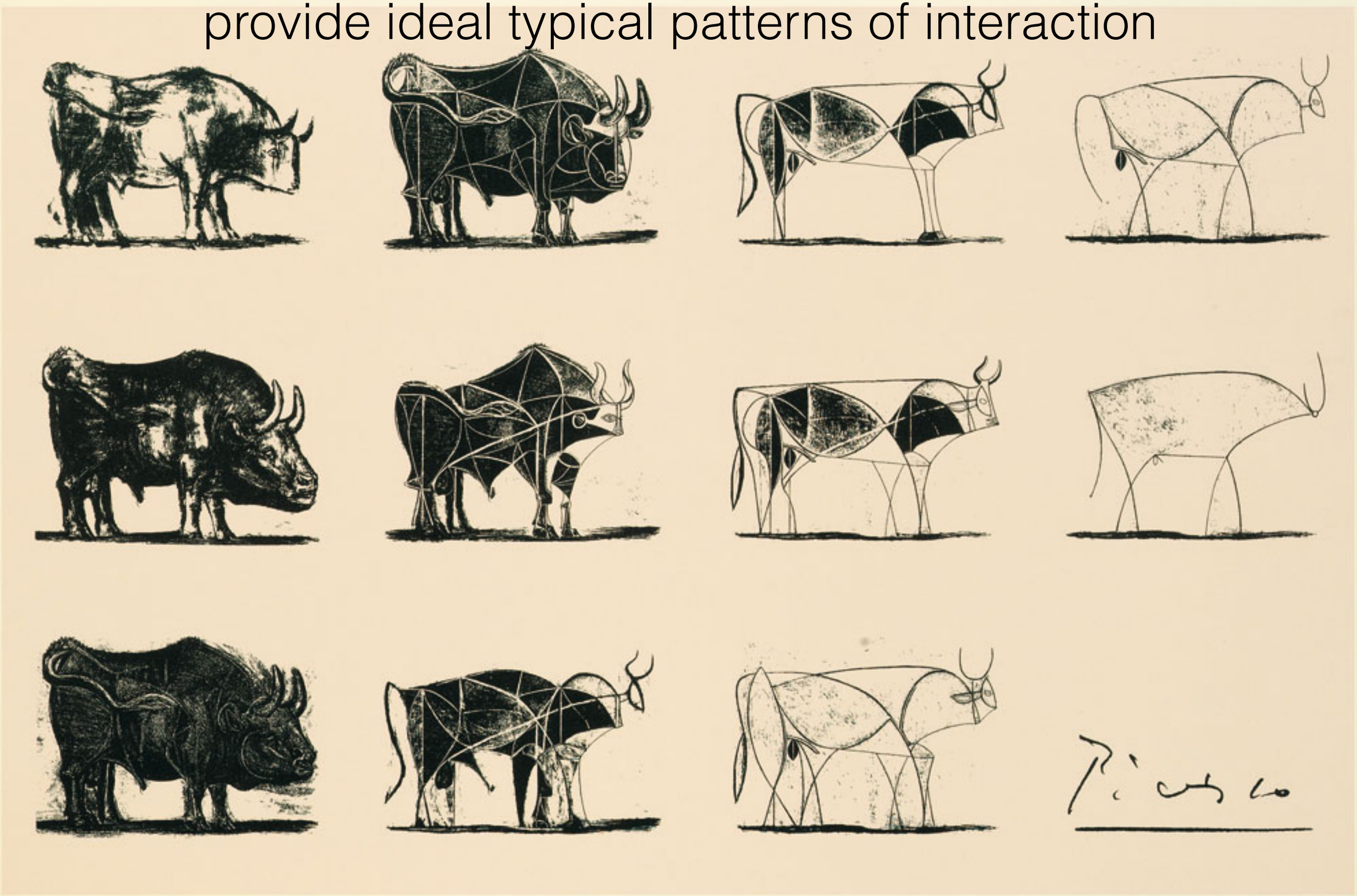
CENTER
AHEAD

# Stylised network models

provide ideal typical patterns of interaction



# Stylised network models

provide ideal typical patterns of interaction

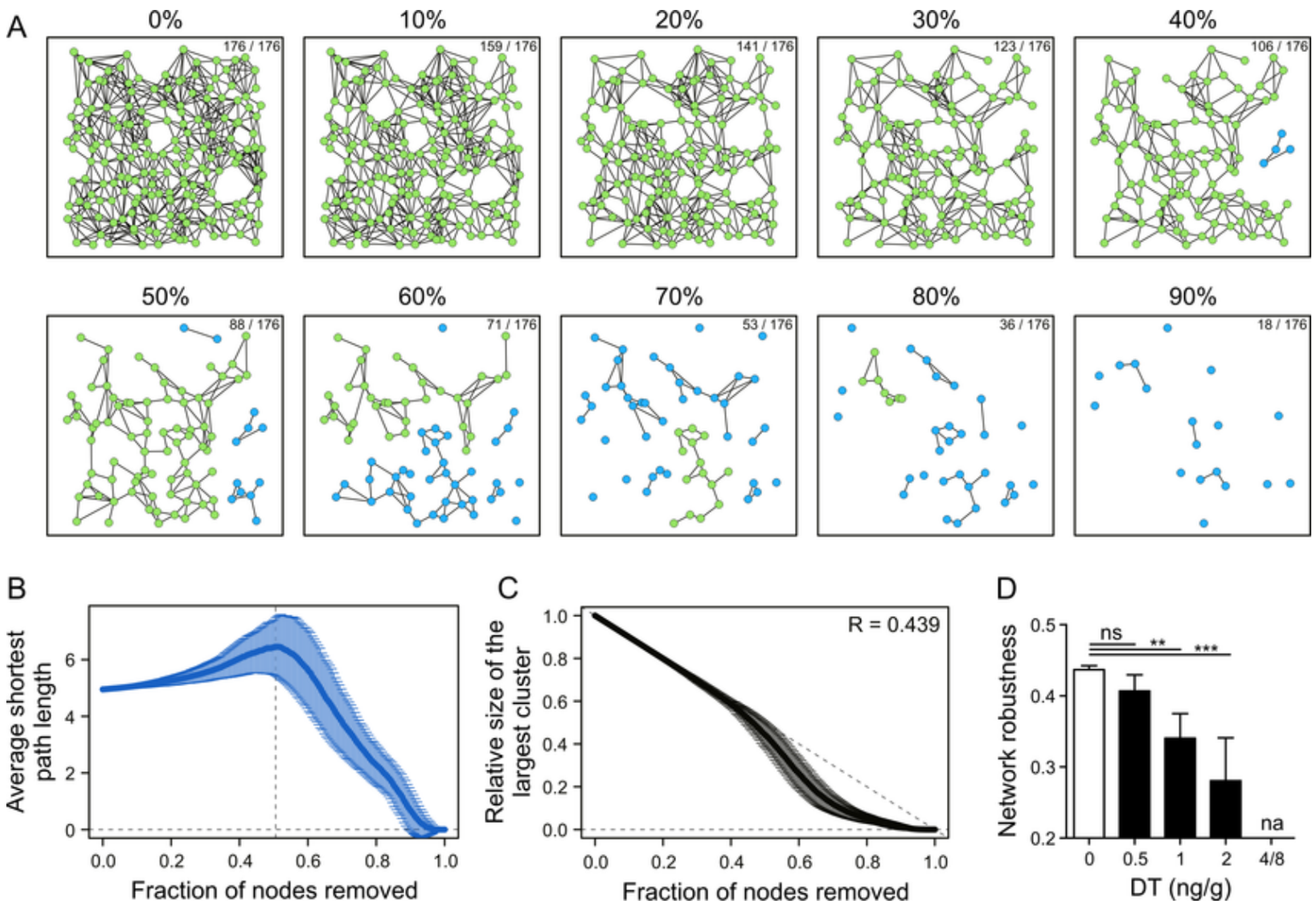
- Can illuminate how micro mechanisms create macro structures
- Good for theory-building and understanding that structure matters

# Stylised network models

provide ideal typical patterns of interaction

- Can illuminate how micro mechanisms create macro structures
- Good for theory-building and understanding that structure matters
- These mechanisms at best capture only part of the story
- Little use in comparing empirical and stylised networks and claim that a “fit” means the network is explained

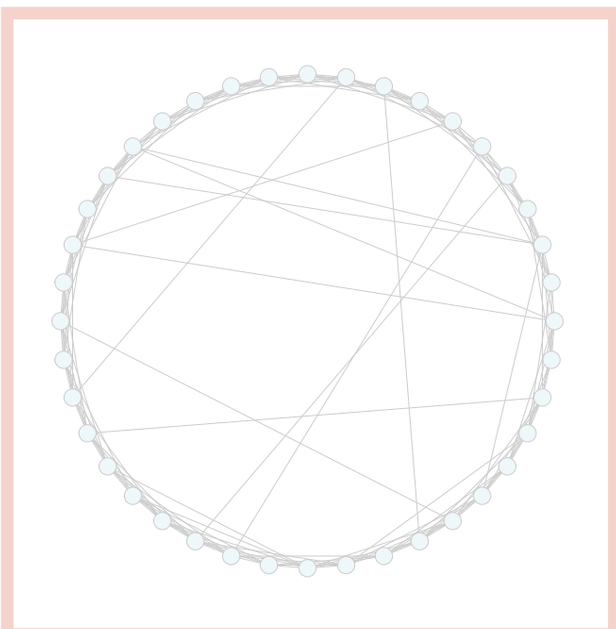
# Network Robustness



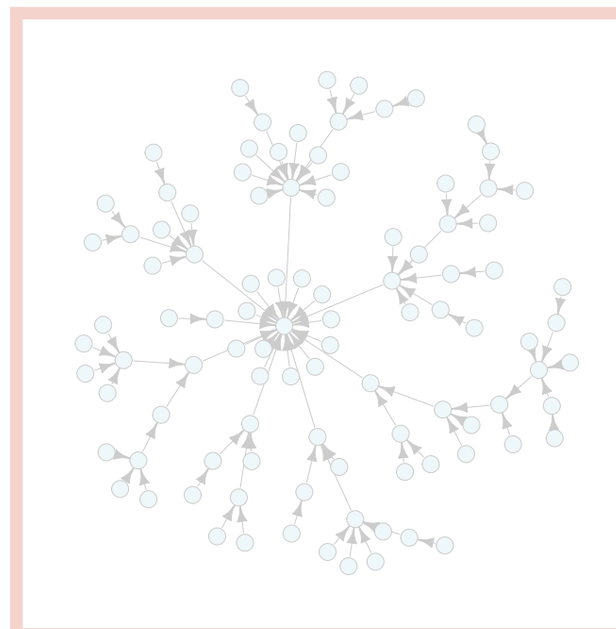

Network percolation theory is a literature that tries to identify how many random or specific nodes can be removed before a network breaks into multiple components

# Topology

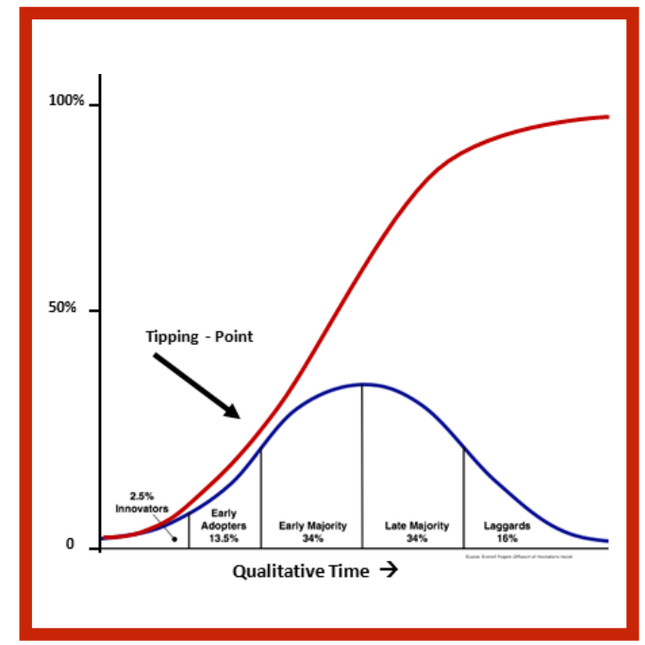
Small-World



Scale-Free



Diffusion

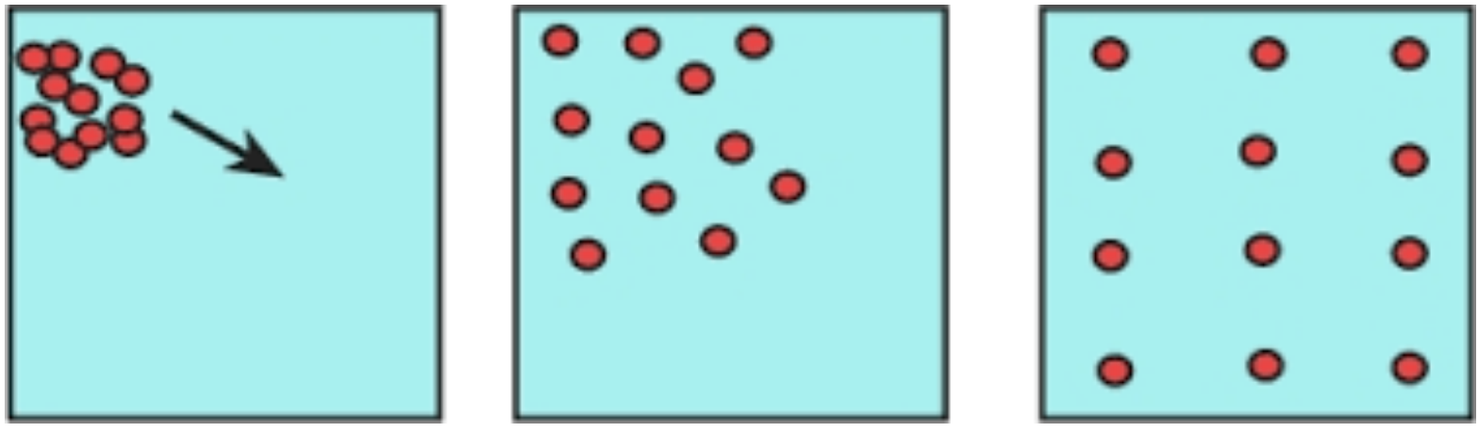


# Networks and Diffusion

- Networks not just abstract structures, also
  - equivalencies for emulation (prisms)
  - conduits for influence (pipes)
- Network ties and structures provide opportunities
- Seek to apply basic ideas of social networks to diffusion

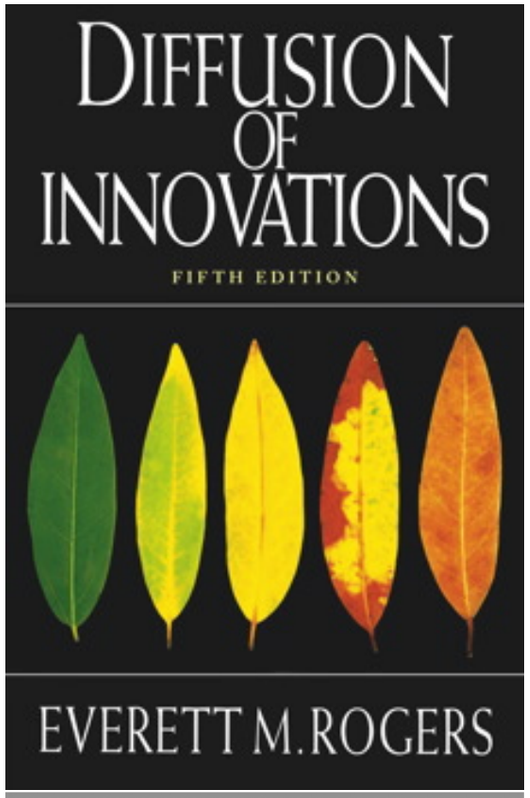
# Diffusion

- According to *Dictionary of the History of Science*, the term ‘diffusion’
  - originated in the 19thC with Graham and Maxwell
  - Latin etymology, *diffundere*, means “to spread out”

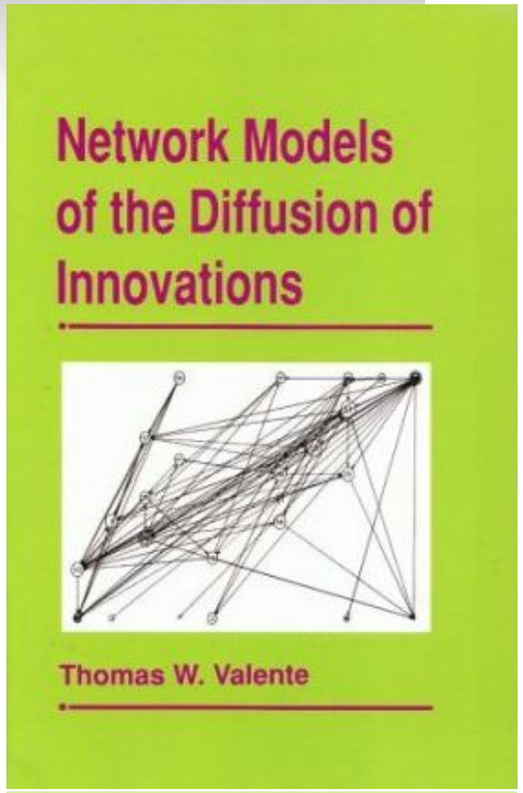


  - Compare with adjacent terms “contagion”, “adoption”, etc.

# Social Diffusion




- Pre-paradigmatic heritage from Galton (cultural traits), Pemberton (postage stamps), and Stuart Chapin (sociology)
- Early diffusion in **rural sociology**
  - 2 rural sociologists at Iowa State University, Ryan and Gross
  - "Diffused" to other midwestern rural sociological researchers in '50s and '60s, and then larger, interdisciplinary field
- By late 1960s, rural sociologists lost interest
  - Not because ineffective scientifically
  - But lack of support during era of overproduction
  - And most interesting questions thought to be answered…
- Now most prevalent in biology: **epidemiology**



# Epidemic Models

- Simulating diffusion across a network involves a basic process:
  - Take a network (may be generated or constructed from data)
  - Select *n* nodes as seeds
  - At each step, each infected node may diffuse its network neighbours with probability $p$ (transmission rate $\beta$)
- This transmission rate can represent a threshold
  - When a balance between acting and not acting is overcome in response to a combination of internal propensities and the influence of external events, such as the perception that "everyone is doing it"
  - Can mean <u>various things</u> in different networks

# Compartmental Models

- Most simple model (Kermack and McKendrick 1927) places nodes into two compartments:
  - Susceptible (S), and
  - Infected (I)
- This model assumes:
  -
  -
  -
  -

# Compartmental Models

- Most simple model (Kermack and McKendrick 1927) places nodes into two compartments:
  - Susceptible (S), and
  - Infected (I)
- This model assumes:
  - Closed population: No birth, death, or mobility
  - Infinite time: Every individual will be infected eventually
  - Homogenous nodes: Each node has same probability to transfer the e.g. disease, innovation, or information to neighbours
  - Homogenous mixing: Or at least random distribution

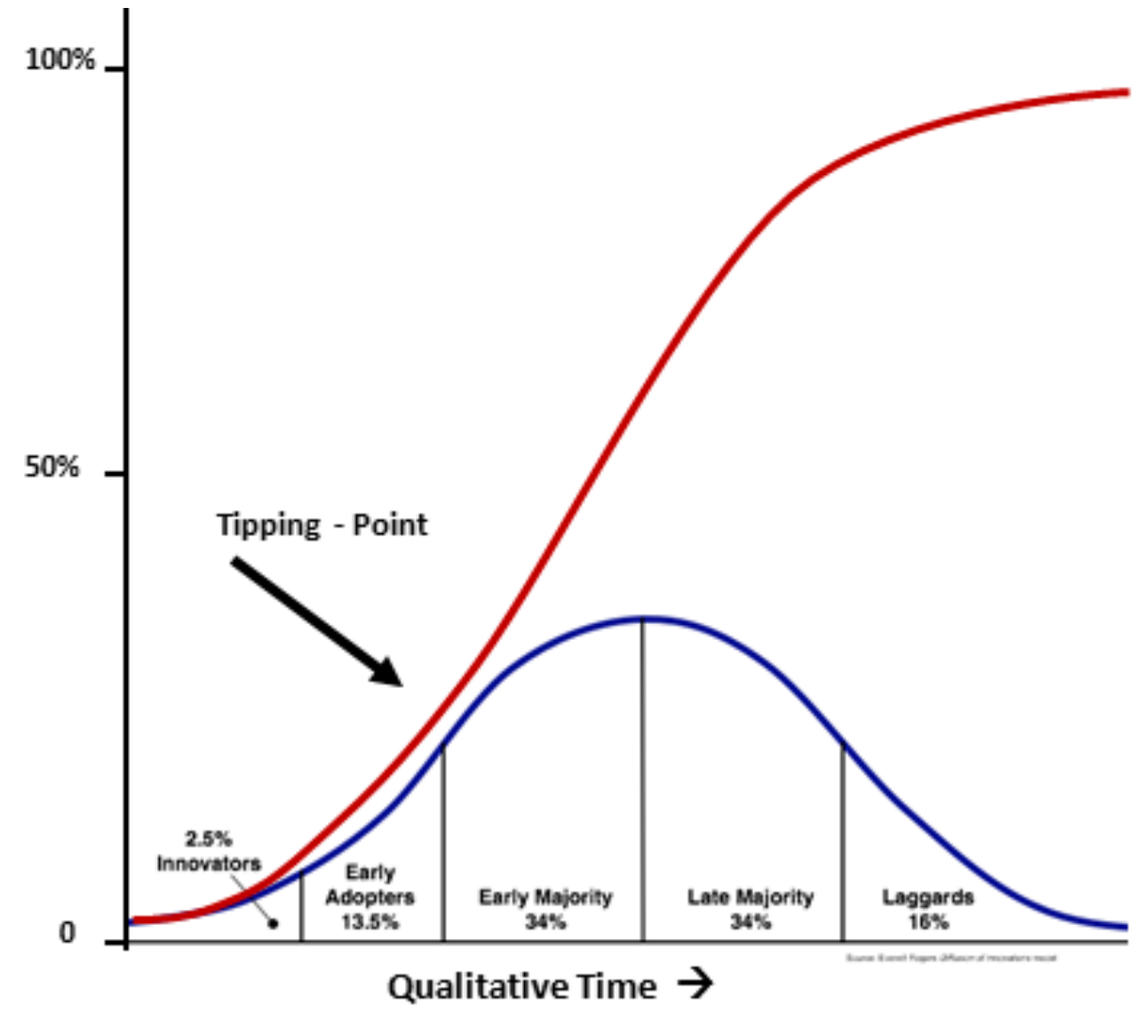
# Diffusion and Adoption

- Given the transmission rate $\beta$, SI model can be described with the differential equations:

$$\frac{dS}{dt} = -\beta SI \qquad \frac{dI}{dt} = \beta SI$$

  - where I + S = 1

- Solving this equation gives us a logistic growth function (an *s*-shaped curve)
  - This curve increases fast after it crosses the critical/tipping point, and grows much slower in the later stage



# Naïveté of the SI Model

- In a real case of epidemic spreading, we have to consider how the status of the infected change…
  - They might recover but become susceptible again (SIS model, like ghonorrea)
  - They might recover and then be immune (SIR model, like chicken pox)
  - They might be removed the network (like death)
- Recovery or other rates can be captured with additional parameters, e.g. $\gamma$, but we will stick with the simpler SI model in this course ;)

# Naïveté of Threshold Models

- *Structurally diverse* social reinforcement
  - Exposure to information from multiple sources has a larger effect when sources independent (unconnected)

- People more likely to join Facebook when invited from friends that are themselves *not connected* (possibly different groups of friends, colleagues, etc) (Ugander et al. 2012)

- Similarly, people more likely to donate to a political campaign when exposed to earlier donors from different groups, rather than from a single group (Traag 2016)

# Naïvete of Diffusion Models

- Vital dynamics: new nodes may enter, with fewer ties and older nodes may perish, taking ties with them

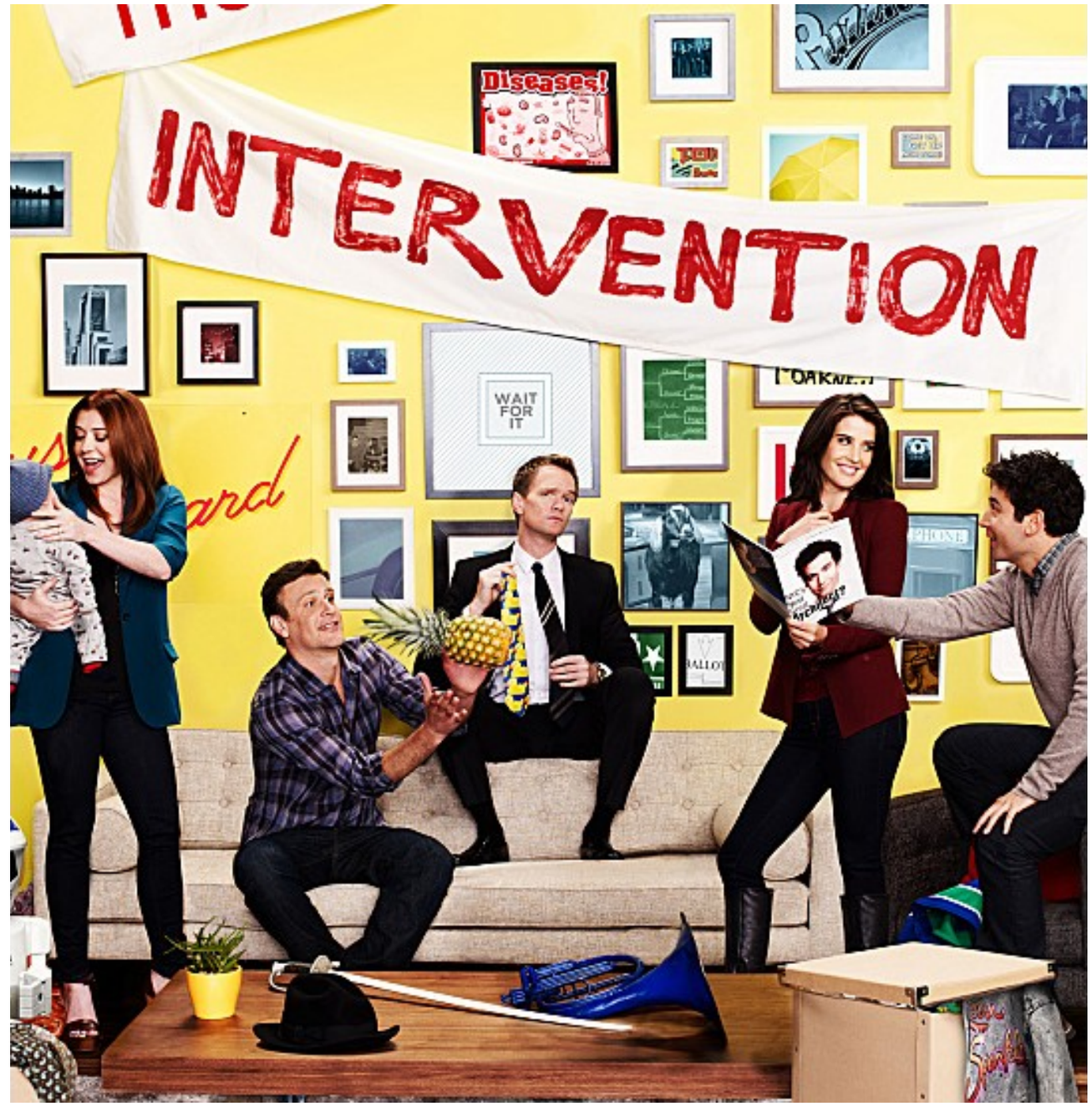
- Serosorting: process by which tie probability is linked to disease status

# More complex models

- http://www.epimodel.org/
  - Disease types include SI, SIR, and SIS epidemics with and without demography, with utilities available for expansion to construct and simulate epidemic models of arbitrary complexity

# Network

Network interventions literature uses network data to assist behavioural change programs by identifying specific nodes to deliver or receive a treatment so as to accelerate or decelerate behaviour change or improve performance




http://vax.herokuapp.com

# Topology

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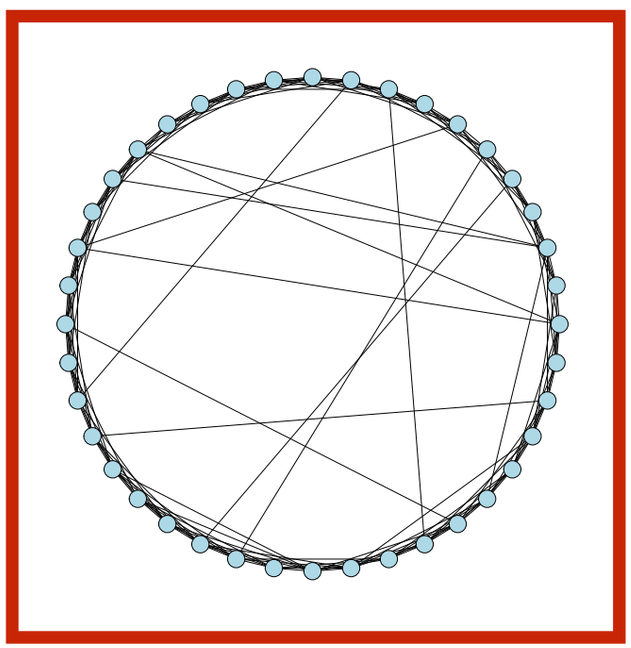

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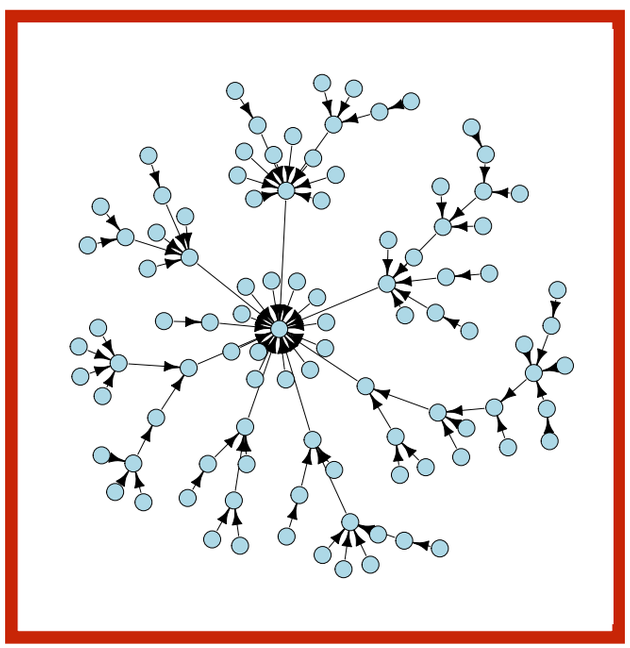
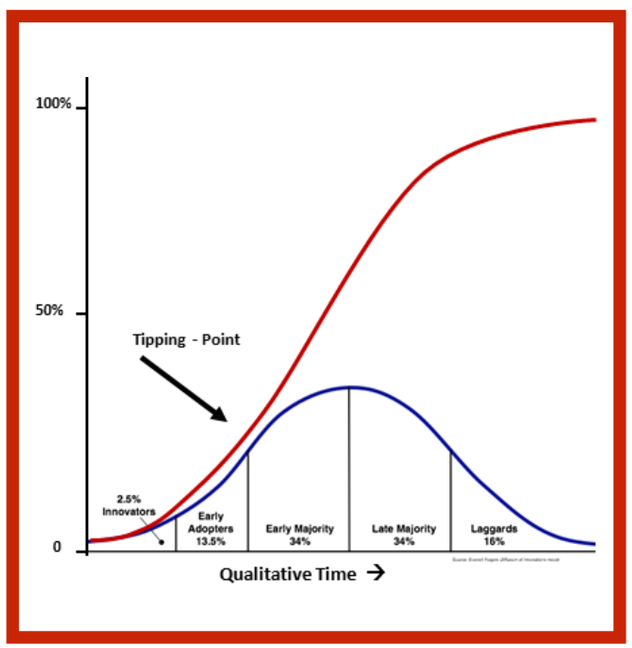
## Diffusion